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B.Tech. Degree III Semester Examination November 2018**ME 1306 MACHINE DRAWING**
(2012 Scheme)

Time : 4 Hours

Maximum Marks : 100

- I. (a) Draw the three views of a hexagonal headed bolt of size M24. The length of the bolt is 80 mm and the thread length is 54 mm. Indicate all dimension and retain all construction lines. (10)
- (b) Draw neat dimensional sketches of the following types of special foundation bolts having diameter 24 mm at the threaded end and indicate all dimensions. (20)
- (i) Lewis type
- (ii) Eye end type (Type A)

OR

- II. Two rods of diameter 20 mm are to be joined using a knuckle pin, collar and taper pin. Prepare the following dimensioned drawings of the joint: (i) Top Half Sectional Elevation (ii) Sectional Plan (iii) End view from left. (30)
- III. An isometric view of a Muff coupling is shown in figure (1). Draw the left half sectional elevation and sectional end view of the coupling. (30)

OR

- IV. Part drawing of a pedestal bearing (Plummer block) is given in figure (2). Assemble them and draw following views: (30)
- (i) Left half sectional elevation
- (ii) Plan left half in section. Also prepare a bill of materials.
- V. Part details of a machine vice are given in figure (3). Draw the following views, all parts assembled in working position (40)
- (i) Full sectional front elevation
- (ii) Plan, lower half in section

OR

- VI. A vertical cross head part details are shown in figure (4), assemble them and draw the following views: (40)
- (i) Left half sectional elevation
- (ii) Plan. Also show a bill of materials.



(P.T.O.)

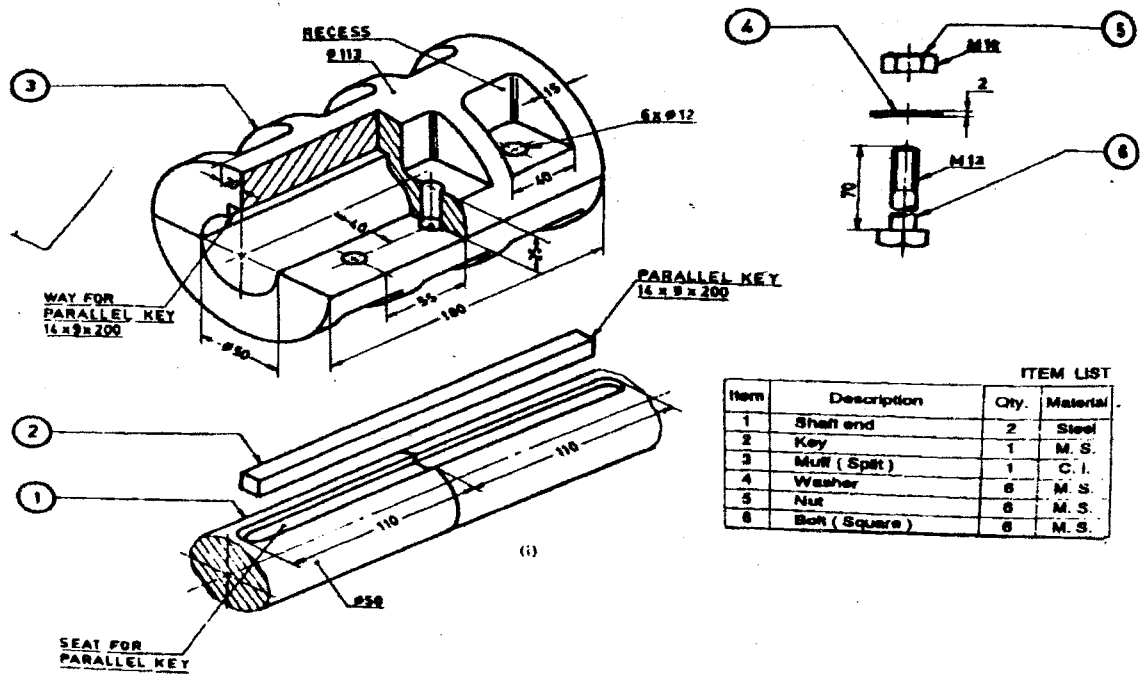


Figure: (1)

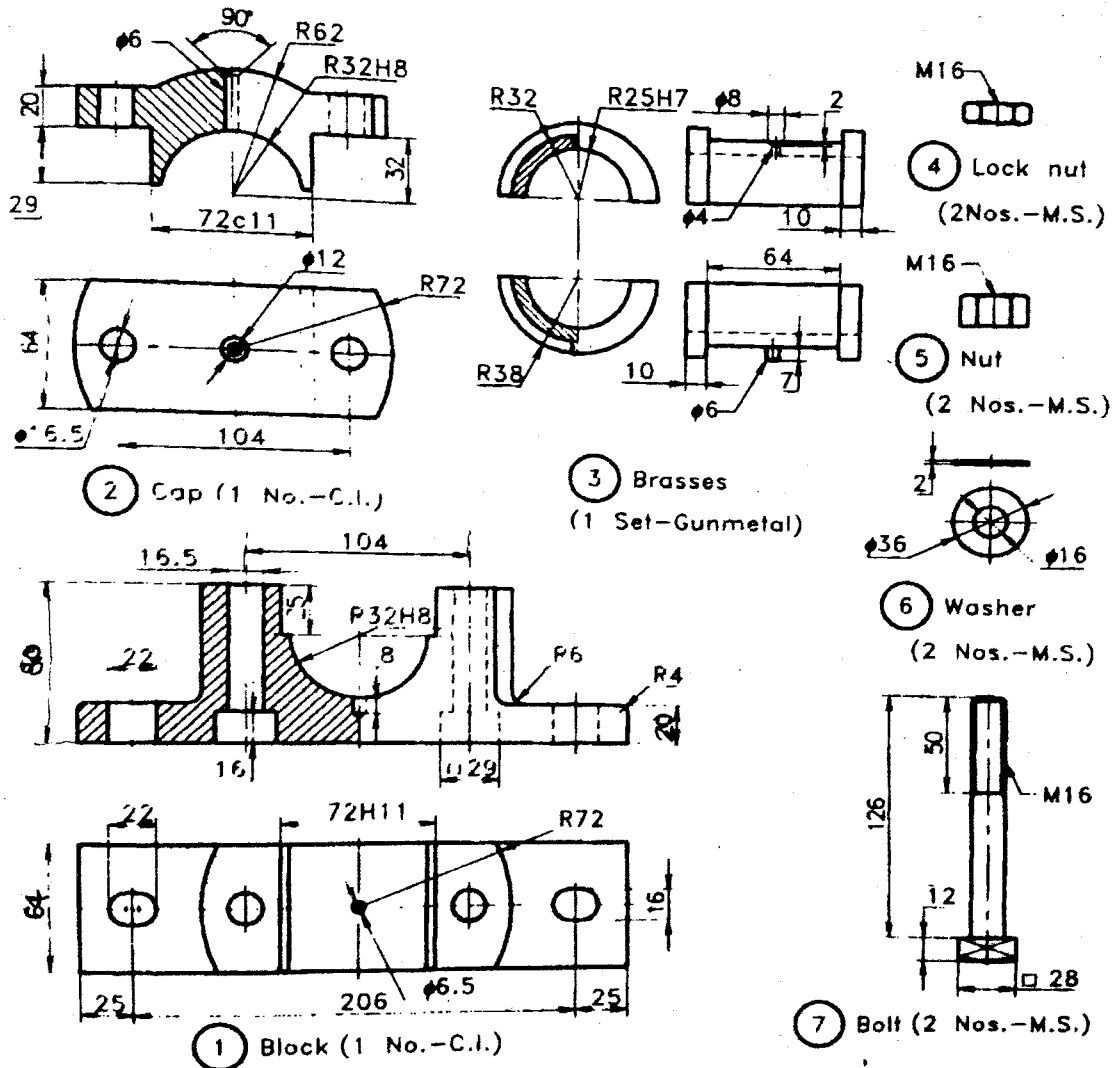
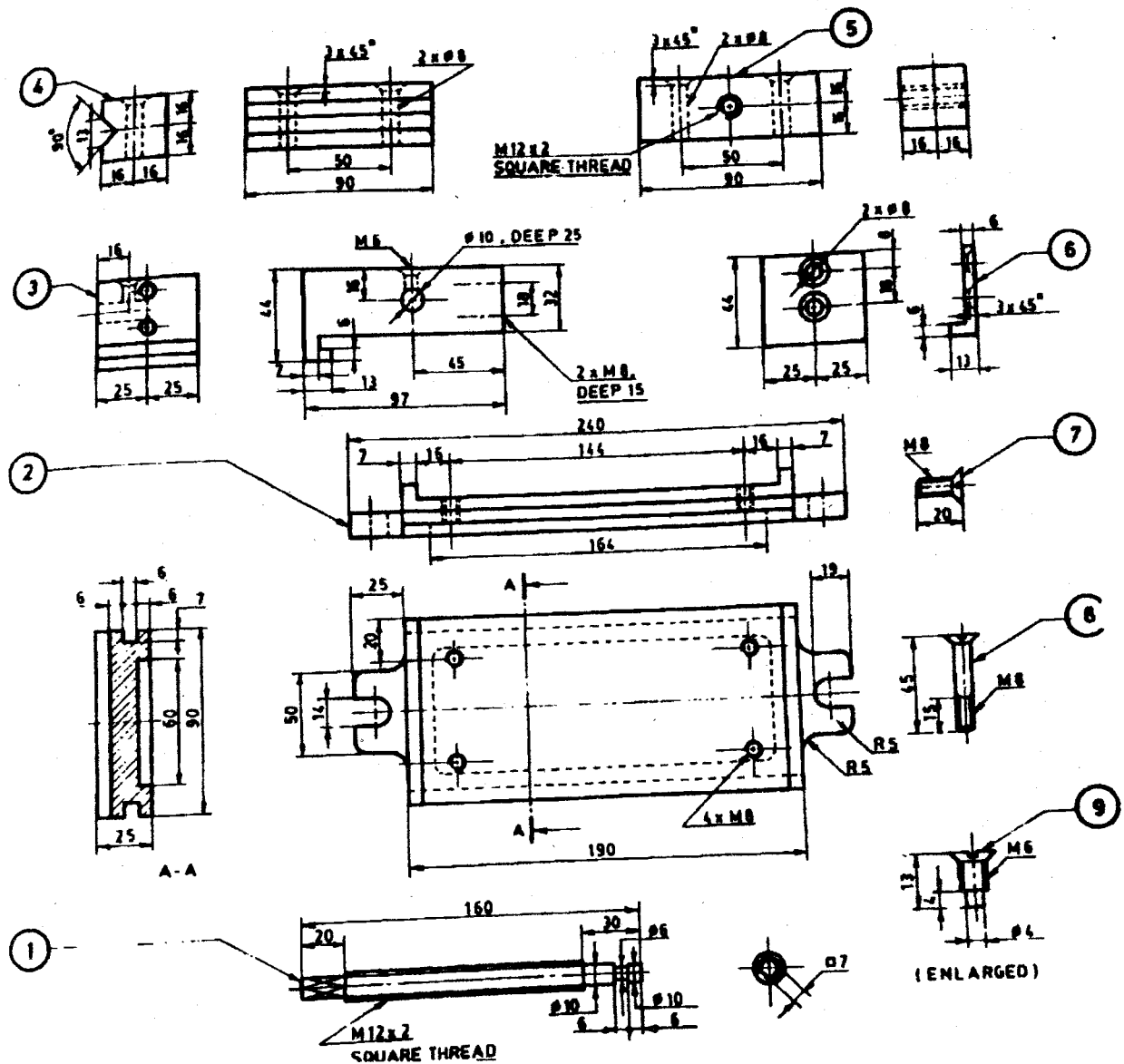


Figure: (2)

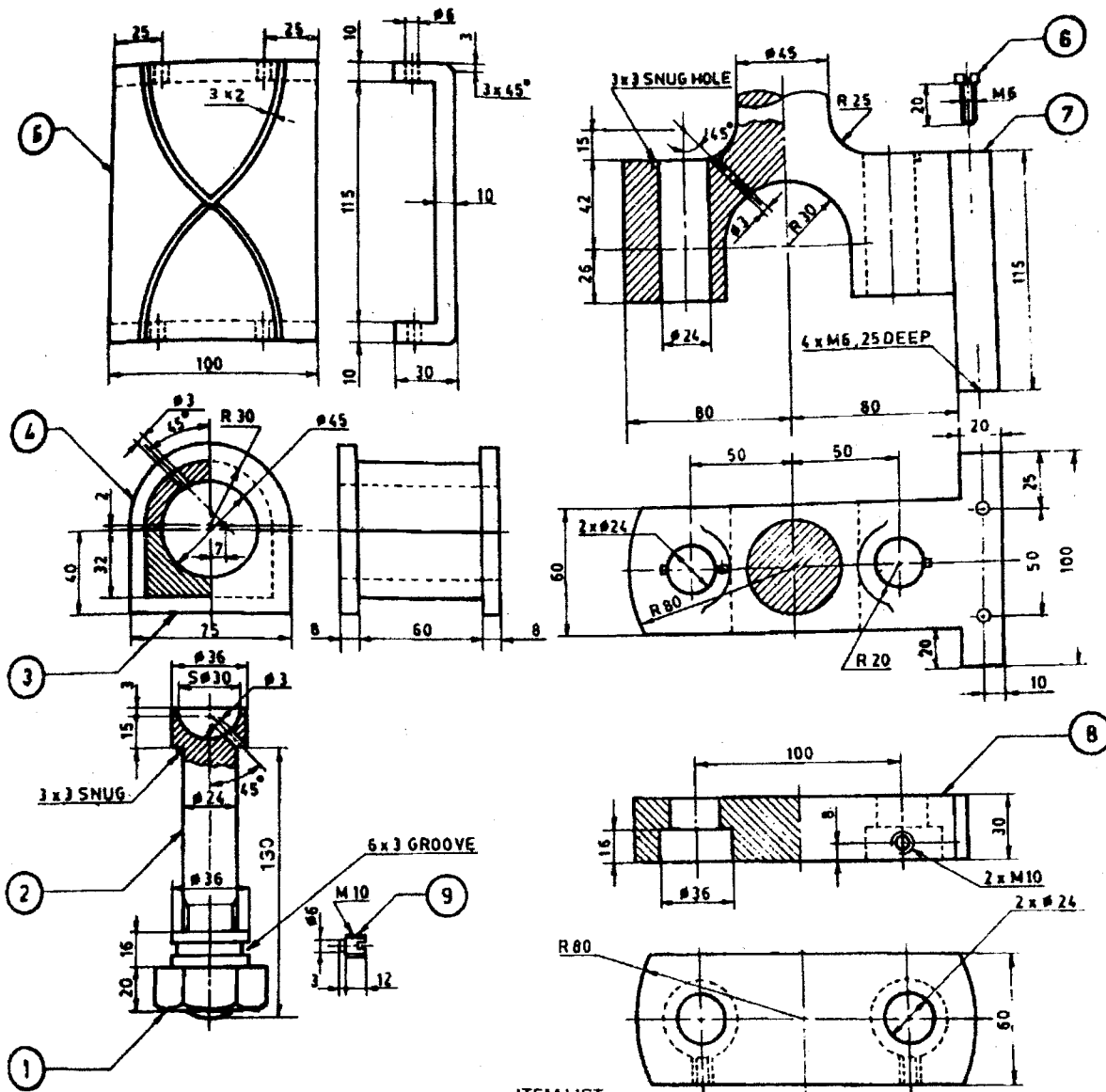


ITEM LIST

Item	Description	Qty.	Material
1	Screw spindle	1	M. S
2	Base	1	C. I.
3	Sliding jaw	1	C. S.
4	Fixed jaw	1	C. S.
5	Block	1	C. I.
6	End plate	1	M. S.
7	Set screw	2	M. S.
8	Screw (M8)	4	M. S.
9	Screw (M6)	2	M. S.

MACHINE VICE

Figure: (3)



ITEM LIST

Item	Description	Qty.	Material
1	Grooved nut	2	M. S
2	Bolt	2	M.S.
3	Bearing (Rectangular)	1	Brass
4	Bearing (Half round)	1	Brass
5	Shoe	1	C.
6	Set screw	4	M S
7	Piston rod end	1	Steel
8	End cap	1	C.S.
9	Grub Screw	2	M S

CROSS HEAD

Figure: (4)

